Fiscal Year:	FY 2024	**Task Last Updated:**	FY 10/09/2023
PI Name:	Rithidech, Kanokporn Ph.D.		
Project Title:	Proteomic Analysis of Mouse Plasma After Space Flight With the Bion-M2 Mission		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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PI Web Page:			
City:	Stony Brook	**State:**	NY
Zip Code:	11794-8691	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology NNH18ZTT002N:Russian Bion-M2 Mission
Start Date:	05/01/2020	**End Date:**	04/30/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:	Bion-M2		
Flight Assignment:	NOTE: End date changed to 4/30/2023 per F. Hernandez/ARC (Ed., 5/3/22) NOTE: End date changed to 4/30/2022 per F. Hernandez/ARC (Ed., 7/27/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Aryal, Uma Ph.D. (Purdue University)		
Grant/Contract No.:	80NSSC20K0994		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	To protect astronauts, we must reduce uncertainties in assessing health risk from space flights and improve our knowledge of space biology to facilitate the future development of biological countermeasures. A deeper understanding of the effects of the space environment on biological systems (i.e., immunological and other physiological conditions) and how the biological systems re-adapt to the 1 g environment are essential. Hence, it is important to conduct studies using omic technologies to discover, at the whole genome level, the cellular/molecular effects of space flight and the mechanisms of re-adaptation to gravity. In this study, we take advantage of the availability of mice to be flown with the Russian Bion-M2 mission. We will use the powerful proteomic technology to characterize protein expression profiles and to determine protein alterations, qualitatively and quantitatively, in plasma samples collected from groups of C57BL/6 mice flown with the Russian Bion-M2 mission at different times post-landing: 1 hour (hr), 12 hr, 5 days (d), and 15 d. Moreover, global and quantitative phosphorylation of proteins will be investigated. The data obtained from these flown mice will be compared to those obtained from two ground control groups: (1) vivarium and (2) flight hardware-housed mice. The results would help to identify proteins that are candidates for functional roles in response to space flight or re-adaptation to the 1 g environment. We will further correlate our proteomic data with the data obtained by the Russian scientists to identify proteins potentially responsible for alterations in biological systems (e.g., immunological and other physiological conditions) in response to space flights and re-adaptation to the 1 g environment. The data will increase our knowledge of space biology and enable the future development of countermeasures. Hence, our findings will provide high-priority and highly relevant information to the Space Biology Element of NASA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Fundamentally, the results from our project will provide information that helps to understand the basic mechanisms that animals use to adapt to the space flight environment, to alterations in gravity and re-adaptation to Earth gravity. Hence, the data will accelerate solutions to biomedical problems affecting human exploration of space as well as human health on Earth.
Task Progress:	Per NASA Ames Research Center, this project has been canceled (Ed., 10/9/2023).
Bibliography Type:	Description: (Last Updated: 01/20/2026)